L&T
5th Sem

CIA - 1
THEORY

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Q1) JavaScript code can be run directly inside a browser without needing a separate compilation step before execution.

Ans:- compilation.
→ Any browser's JavaScript engine reads and runs the code line by line.
→ Hence no separate step to compile is needed

Q2) Write a program to declare a variable 'price' with value 250, then increase it by 50 and print the new value.

Ans: Code -

```
let price = 250;
price = price + 50;
console.log("New Prize:", price);
```

Output:- New Prize : 300

Q3) The value 'undefined' in JavaScript indicates that a variable has been declared but not initialized

Ans:- initialized
→ A value undefined is always given to any variable that is declared using var, let or const. and has no value assigned

Q5) A function that is defined without a name and assigned to a variable is called a anonymous function.

Ans:- anonymous
→ An anonymous function has no name of its own (PTO)

Q4) Write a program to check if two numbers are equal using both == and === and print both results.

Ans :- Program:

```
let a = 5
let b = "5"
console.log ("Using == :", a == b);
console.log ("Using === :" a === b);
```

Output:

```
Using == : true
Using === : false
```

CIA 1 - PRACTICAL

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DS

TASK 1 : Welcome Message.

Code:
```
console.log("Welcome to the Student Record System!");
```

TASK 2: Declare and Update Variables

Code:
```
let systemName = "Student Record System";
let totalStudents = 5;
console.log("System Name:", systemName);
console.log("Total Students:", totalStudents);
```

Output:
```
System Name: Student Record System
Total Students: 5
```

TASK 3: Identify Datatypes

Code:
```
let studentMarks = 85;
let studentName = "Riya";
let isPassed = true;
console.log("studentMarks is a", typeof studentMarks);
console.log("studentName is a", typeof studentName);
console.log("isPassed is a", typeof isPassed);
```

Output:
```
StudentMarks is a number
studentName is a string
isPassed is a boolean.
```

TASK 4: Perform a Calculation

Code:
```
let marks1 = 85;
let marks2 = 78;
let marks3 = 92;
let avg = (marks1 + marks2 + marks3) / 3;
console.log("Avg Marks:", avg);
```

Output: Avg Marks: 85

TASK 5: Classify a Record

Code:
```
let marks = 55;
if(marks >= 40) {
    console.log("Result : Pass");
} else {
    console.log("Result: Fail");
}
```

Output: Result : Pass

TASK 11: Calculate a Derived Value

Code:
```
function getResult (name, marks) {
    if (marks >= 40) {
        return name + " has Passed";
    } else {
        return name + " has Failed";
    }
}
console.log(getResult ("Ankit", 76));
console.log (getResult ("Meera", 32));
```

Output: Ankit has Passed
Meera has Failed.

TASK 12: Arrow Function Version.

Code:
```
const getResult = (name, marks) => {
    if (marks >= 40) {
        return name + " has Passed";
    } else {
        return name + " has Failed";
    }
};
console.log (getResult ("Farhan", 58));
console.log (getResult (" Divya ", 21));
```

Output:

Farhan has passed

Divya has Failed.

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TASK 13: Format Output with Template Literals

Code:

```
let student = {
    name : "Tonya",
    marks : 82,
    section : "B"
};
console.log(`Student ${student.name} from
    Section ${student.section} scored
    ${student.marks} marks.`);
```

Output:

Student Tonya from Section B scored 82 marks.

TASK 14: Track a Date

Code:

```
let admissionDate = new Date(2023, 5, 15);
console.log("Admission Date:" admissionDate.
    toDateString());
```

Output: Admission Date: Thu ~~8~~, Jun 15 2023

TASK 15: Apply a Math Method

Code:

```
let rawMarks = 87.6;
let roundedMarks = Math.round(rawMarks);
let marksList = [72, 88, 65, 91, 79];
let highestMarks = Math.max(... marksList);
console.log("Rounded Marks:", roundedMarks);
console.log("Highest Marks in List:", highestMarks);
```

OUTPUT: Rounded Marks: 88

Highest Marks in List: 91